

Chapter 17 -- Refining Subclasses Through Semantic Differentiation

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17.1 Introduction -- From Taxonomy to Semantics

In Chapter (15) and (16), we focused on building subclass hierarchies inside the **Pizza** ontology.

You learned how subclasses enable:

- semantic specialization
- taxonomy construction
- semantic inheritance
- scalable ontology architecture

At this stage, the ontology already contains an increasingly rich hierarchy.

For example:

```
. /
  |
  └─ Pizza
      |
      └─ NamedPizza
          |
          ├── MargheritaPizza
          ├── AmericanaPizza
          └── SohoPizza
```

This structure provides clear organization already.

However, an important question now emerges:

What actually makes these subclasses semantically different?

At the moment, these subclasses are distinguished mainly by their names.

For humans, names already carry meaning.

We intuitively understand that:

- **MargheritaPizza** typically contains cheese and tomato
- **AmericanaPizza** often includes meat toppings
- vegetarian pizzas exclude meta

But ontology reasoners do not rely on human intuition.

They only understand:

explicit logical statements.

This introduces one of the most important principles in ontology engineering:

If semantic meaning is not explicitly modeled, it does not exist for the reasoner.

This chapter marks a major transition!

We move from:

taxonomy construction

toward:

semantic differentiation.

17.2 Exercise 16 -- Preparing Subclasses for Richer Semantics